

FELLOWSHIP POSTING DESK

OmniPost code map

A guided tour of every package, file, and runtime path.
Where to look, what each piece does, and how the desk,
alarms, captions, and survival layer fit together.

Application `damjay.publicity.omnipost`

Language `Java 17` + XML layouts (no Kotlin)

Store `Room v7` `omnipost.db` + `SharedPreferences`

Alarms `AlarmManager.setExactAndAllowWhileIdle` / `setAlarmClock`

Date `4 September 2026`

Read this before hunting through the tree. Paths are from the repo root.

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1. What this app is

OmniPost is a posting desk for a fellowship (RCCG The Lord's Court youth church). It does not post for you. It nags you, on time, until you write the caption, save it, and mark Posted. The phone is a 4 GB Android with an aggressive OEM battery manager (Tecno / Infinix / Itel / Xiaomi class). If the process dies, the next AlarmClock must still fire at the exact minute.

The product name is OmniPost, never PublicityNag. Package and applicationId are damjay.publicity.omnipost. Source lives under app/src/main/java/damjay/publicity/omnipost/.

What the desk tracks

- Weekly posts: Sunday Service, Wednesday Bible Study, Friday Prayer, plus user-defined weeklies.
- Daily posts and daily countdowns (Beyond Limit until D-Day) as one card, not a pile of days.
- Monthly: fasting tomorrow, fasting today, Happy New Month, birthday NOTICE (end of last month, 10th, 20th).
- Yearly birthdays from the member list.
- Flexible and one-off posts the user adds with a chosen time.

What it will not do

- No generic Android share sheet. WhatsApp is opened with package com.whatsapp only.
- No WorkManager-only scheduling. Exact AlarmClock + START_STICKY FGS + BootReceiver.
- Opening a draft does not count as caption-saved. Only the Save button does.
- The Final Post box is not editable. It shows filled, styled output.

2. How it all works together

Think in four layers that talk in one direction: UI asks the coordinator; the coordinator writes Room and asks the scheduler; the scheduler plants AlarmClocks; the receiver fires sound first, then the coordinator catches up.

```

UI  (Tasks / Settings / Draft / Alarm)
  |
  v
ScheduleCoordinator  bootstrap / onAlarm / markPosted / snooze / saveSeries
  |
  +-- RoutineGenerator      invents Task rows from Series + Members + hardcoded weeklies
  +-- AppDatabase (Room)    tasks, drafts, members, post_series, caption_vars
  +-- AlarmScheduler        setAlarmClock per phase (draft, warning, minute, nag, snooze)
  +-- NagForegroundService  always-on desk notification, START_STICKY, heartbeat
  |
  v
TaskAlarmReceiver  (AlarmManager wakeup)
  |
  +-- AlarmLaunch + AlarmPulse  vibrate/ring NOW from extras (no Room wait)
  +-- AlarmActivity            lock-screen UI, title from extras first
  +-- NagForegroundService.deliverAlarm -> ScheduleCoordinator.onAlarm

```

Process start

OmniPostApp.onCreate (Application) creates notification channels, starts the foreground desk, plants a 60-second heartbeat AlarmClock, and runs ScheduleCoordinator.bootstrap on the disk executor. MainActivity does the same bootstrap again when the user opens the app. BootReceiver does it after reboot, timezone change, time set, and package replace.

One posting cycle

- bootstrap generates missing tasks (occurrenceKey is the unique id of a happening).
- Each active task gets four clocks: draftAt (evening before), warning (default 30 min before post), 1-minute lead, postAt.
- If the caption is saved (task.captionSavedAt > 0), draft and warning clocks are skipped. Status becomes READY until post time.
- At post time the task is NAGGING. The FGS re-bursts on the nag interval until Posted or snoozed.
- Snooze stores snoozeUntilMillis. Next ring is snoozeUntil minus 1 minute (if still in the future), else snoozeUntil.

Threads

AppExecutors.disk() is a single-thread executor for all Room writes. AppExecutors.main() posts to the main loop. Never touch Room on the UI thread. AlarmLaunch and AlarmPulse run on the receiver thread on purpose so sound is not queued behind a database read.

3. Repository layout

Everything that ships is under `app/`. Gradle Kotlin DSL at the root. No Kotlin source.

settings.gradle.kts

Root project name and module include.

build.gradle.kts

Root plugins only.

gradle/libs.versions.toml

Version catalog: AGP 8.13.2, Room 2.6.1, AppCompat 1.7.1, Material 1.13, JUnit 4.13.2.

app/build.gradle.kts

`applicationId`, `minSdk 26`, `targetSdk 35`, `compileSdk 36`, `Java 17`, `viewBinding`, `signingConfigs.omnipost`, `testDebugUnitTest logging`.

keystore/

Release/debug keystore so CI APKs match the phone. Do not rotate it.

.github/workflows/android.yml

On every push: JDK 17, unit tests, `assembleDebug`, upload app-debug APK.

app/src/main/AndroidManifest.xml

Permissions, activities, receivers, FGS `specialUse`, accessibility `keep-alive`.

app/src/main/java/damjay/publicity/omnipost/

All Java. Subpackages: `data`, `notify`, `receiver`, `scheduler`, `service`, `share`, `ui`, `util`.

app/src/main/res/

layouts, values/strings.xml, colors, themes, drawables, mipmap launcher, xml backup/a11y rules.

app/src/test/java/...

JVM unit tests. No instrumented tests in this tree.

docs/OmniPost-code-map.pdf

This file.

Gradle facts

Namespace	<code>damjay.publicity.omnipost</code>
minSdk / target	<code>26 / 35</code>
View binding	On. Layout ids become <code>FragmentSettingsBinding</code> , <code>ActivityAlarmBinding</code> , ...
Signing	Both debug and release use <code>signingConfigs.omnipost</code> (<code>OMNIPOST_STORE_*</code> properties).
Room	<code>runtime + annotationProcessor</code> (Java, not kapt). <code>exportSchema = false</code> .

4. Package map (every Java file)

Root - damjay.publicity.omnipost

OmniPostApp.java

Application. Channels, FGS refresh, heartbeat, bootstrap on disk.

MainActivity.java

Bottom nav: Tasks, Birthdays, Settings. Badge on Settings if survival perms missing. Opens DraftActivity when an intent carries TASK_ID.

data - Room

data/AppDatabase.java

Room database omnipost.db, version 7. Entities: Task, Draft, Member, Series, CaptionVar. Migrations 1-7 are listed and MUST be passed to addMigrations (fallbackToDestructiveMigration would wipe the desk).

data/entity/Task.java

One happening. occurrenceKey unique. seriesId, memberId, captionSavedAt, snoozeUntilMillis, timesLocked, status.

data/entity/Draft.java

Caption box for a task (or free-standing after import). variantA, variantB, finalizedText.

data/entity/Member.java

Birthday person: name, birthMonth, birthDay, notes.

data/entity/Series.java

Repeating template. kind COUNTDOWN / MONTHLY / WEEKLY / DAILY. caption, vars, postHour/Minute, weekdays bitmask, monthly flags, seedKey.

data/entity/CaptionVar.java

Global token {name} with label and value. Unique name index.

data/dao/*Dao.java

Sync getters for the disk thread, LiveData observers for UI. TaskDao also: getActiveSync, getCustomSync, getActiveForSeries, deleteAll. DraftDao: findByTaskId, getAllSync.

scheduler - brain

scheduler/ScheduleCoordinator.java

THE hub. bootstrap, tick, onAlarm, markCaptionSaved, markPosted, reopen, shift, snooze, addCustom, saveSeries, deleteSeries, rememberSeriesCaption, captureLiveCaptions, inheritCaptionSaved, dropStaleCountdowns.

scheduler/RoutineGenerator.java

Pure Java. Turns now + members + series into a list of Task candidates. Hardcoded weeklies and monthlies plus addSeries for user templates.

scheduler/AlarmScheduler.java

AlarmManager. Phases 1-7. setAlarmClock with show intent to MainActivity. cancelTask cancels every phase for that id. Heartbeat, kick (5s), watchdog 5:00 AM.

scheduler/TaskStatus.java

SCHEDULED, DRAFTING, READY, WARNING, NAGGING, SNOOZED, POSTED. dueStatus, captionIsSaved, nextRingMillis, nextToRing (soonest clock wins, even if snoozed).

scheduler/TaskTypes.java

Type strings, labels, cadence, section, isCustom, oneCard (countdown / notice / daily), seriesGroup.

scheduler/TaskSections.java

Home grouping: needs-you first, then next 48 hours, then weekly / monthly / campaign.

scheduler/CaptionTemplates.java

Fill {tokens}. isCanned, lingerDraft, sharedTemplate, seriesOf, countdownDays from occurrenceKey.

scheduler/CaptionVars.java

Inbuilt token list, reserved names, nested expand, {range:today:date}, {today+3}.

scheduler/SeriesDefaults.java

Seed series beyond-limit and birthday-notice, canned caption bodies, blank templates.

scheduler/Campaigns.java

Beyond Limit 9-13 September 2026, BIRTHDAY_WA link.

scheduler/ScheduleTimes.java

All durations and default hours in one place (30 min warning, 1 min lead, 5 min ring, 30 s escalate vibrate, heartbeat 60 s, generate 3 countdown days).

scheduler/DateUtils.java

nextWeekdayAt, dayKey, prettyClock/Date/Range, nagFromOrigin, formatUntil, formatStamp.

scheduler/Weekdays.java

Bitmask of Calendar weekdays, sentence(), clock().

notify - sound and shade

notify/AlarmLaunch.java

fromIntent / fire. Uses extras (TASK_TITLE, POST_AT), not Room. Starts pulse, AlarmActivity, cue notification.

notify/AlarmPulse.java

Media + vibrator. begin (loud) vs beginSoft (caption write). Escalate: vibrate 30 s then ring 5 min. silence() + generation so a new pulse does not resurrect a stopped one. isQuiet after hush.

notify/AlertPlan.java

Pure durations from alert mode. No Android types so tests can read them.

notify/Alerts.java

Low-level sound/vibrator helpers used by AlarmPulse.

notify/NotificationHelper.java

Channels draft / alarm.v3 / ongoing.v2. buildDesk (FGS), notifyCue, hush, cancelForTask, fullScreen PendingIntent to AlarmActivity.

receiver

receiver/TaskAlarmReceiver.java

AlarmClock entry. WakeLock 15 s. AlarmLaunch first, then FGS.deliverAlarm. Do not add Room here.

receiver/BootReceiver.java

BOOT_COMPLETED, QUICKBOOT, MY_PACKAGE_REPLACED, TIMEZONE_CHANGED, TIME_SET. goAsync + bootstrap.

receiver/MarkPostedReceiver.java

Notification action 'Mark posted'.

service

service/NagForegroundService.java

START_STICKY specialUse FGS. Desk notification is the one that will ring first. onStartCommand: optional AlarmLaunch, then onAlarm or tick. Burst nags on interval. onTaskRemoved / onDestroy schedule kick+heartbeat and restart.

service/DeskKeepAliveService.java

AccessibilityService that ignores events. Exists only so OEM killers treat OmniPost as a keep-alive app. Does not read the screen.

ui

ui/TasksFragment.java

Home list. Filter pending/posted. FAB add weekly/daily/custom/countdown/monthly. Long-press / edit series via CaptionTemplates.seriesOf.

ui/TaskAdapter.java

Cards, chips, grouping headers (item_day_header).

ui/BirthdaysFragment.java

Member list, add/edit dialog.

ui/MemberAdapter.java

Birthday rows.

ui/SettingsFragment.java

Alert mode, nag interval, warning, draft hour, seed captions, desk ongoing, fullscreen, variables, series list, export/import, survival rows, 70 s test, re-arm.

ui/DraftActivity.java

Write caption. A/B side by side (landscape both, portrait swipe). Final Post preview. persist(false) on pause; persist(true) on Save marks caption saved. rememberSeriesCaption with raw A.

ui/DraftsFragment.java + DraftAdapter.java

Draft list + delete. Not on the bottom nav; captions are written from the task. Still compiled.

ui/AlarmActivity.java

Lock-screen alarm. Paints title/phase from extras immediately, then load() fills from Room. Any action calls acknowledge() -> hush. Snooze labels are clock times (Nag me by 10:30 AM).

ui/SnoozeChooser.java

30 min / 1 h / 3 h / tonight / tomorrow / pick datetime. nagFromOrigin.

ui/SeriesEditor.java

Dialog for repeating posts (dialog_series.xml).

ui/VariablesActivity.java + VariableEditActivity.java + VarAdapter.java

Caption variables screen. Inbuilt first, then user tokens.

share

share/WhatsAppRouter.java

Clipboard + ACTION_SEND with setPackage(com.whatsapp), then w4b. Never Intent.createChooser.

share/WhatsAppPreview.java

Preview helper for WhatsApp-style markup.

share/InstagramStyle.java

bold -> mathematical bold, *_italic_* / enclosed dashes -> italic, both -> bold italic Unicode.

util

util/Prefs.java

omnipost.prefs: alert_mode, nag_minutes, warning_minutes, draft_hour, seed_captions, desk_ongoing, full_screen, last_nag_burst, series_defaults_v1.

util/DeskBackup.java

JSON snapshot encode/decode/collect/apply. Format omnipost-backup v1.

util/AppExecutors.java

disk single thread + main Handler.

util/ExtraKeys.java

Intent extras: TASK_ID, DRAFT_ID, PHASE, PULSE, VAR_ID, TASK_TITLE, POST_AT.

util/SurvivalHelper.java

Checks and settings deep-links for notifications, exact alarms, battery, fullscreen, overlay, accessibility.

5. Data model (Room + prefs)

tasks

A Task is one ringing. Repeating work is not stored as a recurrence rule on the task; RoutineGenerator inserts the next few happenings, keyed by occurrenceKey so bootstrap can upsert without duplicates.

- occurrenceKey examples: SUNDAY_SERVICE|2026-09-05, COUNTDOWN|2026-09-09|2026-09-04, WEEKLY|{seriesId}|{day}, BIRTHDAY|{memberId}|{day}, ONE_OFF|{postAt}|{titleHash}.
- Countdown keys: 3-part COUNTDOWN|end|postDay or 4-part COUNTDOWN|{id}|end|postDay. countdownDays = daysBetweenKeys(post, end). 4 Sep -> 9 Sep = 5.
- captionSavedAt > 0 means the user hit Save. Status then skips DRAFTING/WARNING.
- timesLocked: user shifted the post time; generator must not overwrite draftAt/postAt.
- oneCard types (countdown, birthday notice, daily) show as a single row. markCaptionSaved copies captionSavedAt to active siblings of that series.

post_series

User-editable repeating templates. seedKey 'beyond-limit' and 'birthday-notice' mark the two built-ins so bootstrap can backfill without duplicating. Custom series clear seedKey on save. caption is the lingering template; vars are name=value lines local to that series (they win over global caption_vars).

drafts

Linked by taskId. Opening DraftActivity may insert a seed row; that must not set captionSavedAt. After uninstall-restore, imported drafts have taskId 0 and still appear as saved words.

members / caption_vars

Members drive yearly BIRTHDAY tasks. Caption vars are global {theme}, {venue}, ... Values may themselves contain tokens.

Migrations (do not skip addMigrations)

- 1->2 tasks.snoozeUntilMillis
- 2->3 tasks.timesLocked
- 3->4 tasks.seriesId + table post_series
- 4->5 post_series.endAtMillis, vars
- 5->6 post_series.weekdays + table caption_vars
- 6->7 tasks.captionSavedAt

Prefs (not Room)

Alert mode escalate|both|sound|vibrate. Default escalate = vibrate 30 s then ring 5 min. Nag every 5 min. Warning 30 min. Draft hour 20:00 the evening before. seedCaptions default false (weeklies start blank). deskOngoing default true. fullScreen default true. series_defaults_v1 remembers that Beyond Limit + NOTICE were installed.

6. The alarm and survival pipeline

Why this shape

OEM killers drop background work. WorkManager is not good enough. The contract is: AlarmManager.setAlarmClock (and setExactAndAllowWhileIdle as fallback), a foreground service that returns START_STICKY, and BootReceiver to re-plant everything after reboot.

Phases (AlarmScheduler)

PHASE_DRAFT = 1	draftAtMillis. Soft vibrate. Write the caption.
PHASE_WARNING = 2	postAt minus warning lead (default 30 min). Soft. Caption-ready.
PHASE_NAG = 3	postAt. Loud. Keep coming until Posted.
PHASE_PULSE = 4	Heartbeat / FGS pulse. taskId 0 = global tick.
PHASE_WATCHDOG = 5	05:00 daily. Full bootstrap.
PHASE_SNOOZE = 6	Wake from snoozeUntil. Loud like nag.
PHASE_MINUTE = 7	postAt minus 60 s (also snoozeUntil minus 60 s). Loud. Send it now.

Sound before UI

TaskAlarmReceiver must not wait for Room. Title and postAt ride on the PendingIntent extras (ExtraKeys.TASK_TITLE, POST_AT) so AlarmActivity can paint the heading on the first frame. AlarmPulse starts in the same onReceive. Then FGS is started so the process stays promoted after the 10-second receiver window.

Escalate mode

Default. Vibrate 30 seconds; if the user has not tapped Open caption / I posted it / Flyer not ready, ring for 5 full minutes. Tapping any alarm action calls NotificationHelper.hush -> AlarmPulse.silence. isQuiet must stay set so the FGS burst does not start the ring again.

Soft vs loud

Writing the caption is not as intense as sending the flyer. PHASE_DRAFT and PHASE_WARNING use beginSoft (short vibrate, ScheduleTimes.GENTLE_MS). PHASE_MINUTE, PHASE_NAG, PHASE_SNOOZE are loud.

Foreground desk

NagForegroundService shows 'Next on the desk' using TaskStatus.nextToRing - the soonest clock, not the original post time. A fasting card snoozed until 5:00 PM must beat a birthday NOTICE snoozed until 7:30 PM. If deskOngoing is on, the service never self-stops. Swipe-away schedules a 5-second kick AlarmClock and restarts the service.

Survival settings

Settings -> Stay alive on this phone. Rows: notifications, exact alarms, unrestricted battery, full-screen intents, display over other apps, keep-alive (accessibility). Tecno/Infinix copy tells the user to enable Autostart. The 70-second test inserts a TEST task (draft in 5 s, nag in 70 s) so they can swipe OmniPost away and prove the engine survived.

7. Generating the desk

ScheduleCoordinator.bootstrap is the only place that rebuilds the world. Order matters:

- Delete stale TEST tasks older than 24 h and cancel their alarms.
- captureLiveCaptions: for each active live task, if the draft is custom and the series caption is still canned, copy draft.variantA onto the series.
- dropStaleCountdowns: countdown tasks whose post day is before today are deleted (after rememberTaskDraft) so 9-days / 8-days / 7-days never stack.
- ensureSeriesDefaults: insert Beyond Limit + birthday NOTICE once (Prefs.seriesDefaultsInstalled). Backfill empty vars/endTime on existing seeds.
- RoutineGenerator.generate(now, tz, members, enabled series, draftHour).
- For each candidate: if no row with that occurrenceKey, inheritCaptionSaved from a sibling (oneCard) and insert; if exists and not POSTED, update title/description/seriesId and times unless timesLocked.
- applyDueAndSchedule: recompute status, fire transitions (showDraft / showWarning / showNagBurst), AlarmScheduler.scheduleTask each.

What RoutineGenerator emits

- Hardcoded weeklies: Saturday 10:00 Sunday Service, Wednesday 10:00 Bible Study, Friday 10:00 Prayer. Four weeks ahead.
- Month end 07:00 Fasting tomorrow; month start 07:00 Fasting today AND Happy New Month. Two months ahead.
- Each enabled Series: weekly (bitmask days), daily (one card per day, GENERATE_WEEKLY_COUNT window), countdown (GENERATE_COUNTDOWN_DAYS = 3 rolling days until D-Day), monthly notice (EVE / D10 / D20).
- Each Member: next birthday 07:00.

Draft time is always the evening before at Prefs.draftHour (default 20:00), not 10:00 on the day before. That is why Wednesday-for-tomorrow must not ring in the morning of the day before.

Coordinator actions the UI calls

markCaptionSaved	captionSavedAt = now, status = dueStatus (READY if still before post). Copies to oneCard siblings. hush + reschedule.
rememberSeriesCaption	Writes template to series.caption unless isCanned. Also used from DraftActivity persist.
markPosted / reopen / shift / snooze	Shows + alarms + FGS refresh.
addCustom / deleteCustom	FLEXIBLE or ONE_OFF with timesLocked.
saveSeries / deleteSeries	Cancel that series' tasks, then bootstrap.

8. Captions, tokens, linger, and Save

Two different 'saved' ideas

The words live on `Series.caption` (the template for every later day) and on `Draft.variantA` (this card). The nag state lives on `Task.captionSavedAt`. Only `DraftActivity.persist(true)` - the Save button - with a non-empty pick and `taskId > 0` calls `markCaptionSaved`. `onPause persist(false)` may remember the series template but must not mark. Opening Drafts / auto-inserting a seed must not set `DRAFTING` if the caption is already saved, and must not count as saved.

Linger (countdown / notice / weekly / daily)

Write once, reuse every later day, fill `{Days}/{away}/{month}` at preview time. Never refill `SeriesDefaults.countdownCaption()` over a custom body.

- `isCanned(text)` is true for empty, the Beyond Limit seed, the NOTICE seed, and the blank templates.
- `rememberSeriesCaption` skips canned, so autosave of a seed cannot clobber a custom series caption.
- `sharedTemplate`: custom series caption, else sibling, else seed. `DraftActivity` seeds from series only.
- `lingerDraft`: if the box is still canned and the shared template is custom, upgrade the box. Never overwrite a custom box.
- `load()` must not overwrite `variantA` with a filled (token-replaced) string. Fill happens in the Final Post preview only.
- `seriesOf`: `task.seriesId`, else `findBySeed` for countdown / NOTICE.

Fill pipeline

```
CaptionTemplates.apply(source, task, series, extras)
-> parse series.vars (series wins)
-> extras = CaptionVars.map(db) global tokens
-> putBuiltins {today, date, weekday, month, Days, away, days, range, ...}
-> CaptionVars.expand (up to 8 passes: nested values, {today+3}, {range:today:date})
```

`{date}` is the proposed due day. `{today}` is the calendar day on the phone right now. They are not the same. Birthday NOTICE `{month}` comes from the `occurrenceKey` month slot (the month being celebrated), not necessarily today's month.

Inbuilt tokens (Variables screen, first section)

`today`, `today_weekday`, `today_month`, `today_year`, `now`, `clock`, `date`, `due`, `weekday`, `month`, `year`, `start`, `end`, `range`, `Days`, `days`, `away`, `name`. Plus composed `{range:today:date}` and offsets `{today+3}` `{date-1}`.

A/B and Final Post

`DraftActivity` has Variation A and B. Portrait: horizontal pager, one column at a time. Landscape: both. Final Post is read-only and shows `CaptionTemplates.apply` then Instagram-style preview of what will actually look like (bold/italic already applied in that box as WhatsApp markup still visible; Copy for Instagram runs `InstagramStyle.toUnicode`).

9. Screens and layouts

Bottom navigation (res/menu/bottom_nav.xml)

- Tasks -> TasksFragment (fragment_tasks.xml, item_task.xml, item_day_header.xml)
- Birthdays -> BirthdaysFragment (fragment_birthdays.xml, item_member.xml, dialog_member.xml)
- Settings -> SettingsFragment (fragment_settings.xml, item_setting_row, item_switch_row, item_survive_row)

Activities

DraftActivity	activity_draft.xml. Parent MainActivity. Extra TASK_ID or DRAFT_ID.
AlarmActivity	activity_alarm.xml. Theme AppTheme.Alarm, showWhenLocked, turnScreenOn, excludeFromRecents, no animation. Extras TASK_ID, PHASE, TASK_TITLE, POST_AT.
VariablesActivity	activity_variables.xml. List of inbuilt + user vars.
VariableEditActivity	activity_variable_edit.xml. Name, label, value, live preview.

Dialogs

layout/dialog_series.xml

SeriesEditor: name, kind, days, time, caption, vars, D-Day / monthly slots.

layout/dialog_custom_task.xml

TasksFragment FAB: title, when, flexible vs one-off, time-remaining before Save.

layout/dialog_member.xml

Name + month + day.

Strings and theme

res/values/strings.xml

All user-visible copy. Keep the voice: short, desk, nag, caption.

res/values/colors.xml + themes.xml

Navy / cream / gold. values-night variants exist.

res/mipmap-anydpi-v26/ic_launcher*.xml

Adaptive icons. Drawables under res/drawable and mipmap densities.

Home ordering

TaskSections + TaskAdapter: items that need you (NAGGING, WARNING, DRAFTING, SNOOZED) first, sorted by nextRingMillis, then the next 48 hours, then weekly / monthly / campaign groups. The first row is the soonest to ring.

10. Sharing to WhatsApp and Instagram

DraftActivity: Send to Important Info Peeps and Final Post both call `WhatsAppRouter.sendExplicit`. That copies the filled caption, then `ACTION_SEND text/plain` with `setPackage(com.whatsapp)`, falling back to `com.whatsapp.w4b`. Manifest `<queries>` lists those two packages. If WhatsApp is missing, the caption is already on the clipboard and a toast says so.

Copy for Instagram runs `InstagramStyle.toUnicode` on the filled caption: asterisks become bold mathematical letters, underscores (and enclosed dashes) become italics, nested `*_both_*` become bold-italic. WhatsApp markup must not leak into the Instagram paste.

11. Export / import (survive uninstall)

Settings -> Keep a copy. SAF CreateDocument / OpenDocument so the file lives in Drive or Downloads, not inside the app sandbox (uninstall would delete that).

util/DeskBackup.java

collect / apply / encode / decode. JSON format omnipost-backup, version 1.

The file contains:

- prefs: alertMode, nagMinutes, warningMinutes, draftHour, seedCaptions, deskOngoing, fullScreen
- series: every repeating template including captions and times
- vars: caption variables
- members: birthdays
- drafts: A/B/final text (taskId cleared on import)
- customTasks: non-posted FLEXIBLE and ONE_OFF

Import confirms, then cancels alarms, replaces those tables, writes prefs, sets seriesDefaultsInstalled, and bootstrap() so clocks come back. Generated weeklies/monthlies/birthdays are recreated from members + series, not snapshotted row-for-row (except custom tasks).

Android backup_rules.xml / data_extraction_rules.xml also include the db and prefs for cloud backup, but OEM backup is unreliable - the explicit file is the real safety net.

12. Build, signing, and CI

`.github/workflows/android.yml`

push, pull_request, workflow_dispatch. testDebugUnitTest then assembleDebug. Artifact name app-debug.

Signing: app/build.gradle.kts signingConfigs.omnipost reads OMNIPOST_STORE_FILE, OMNIPOST_STORE_PASSWORD, OMNIPOST_KEY_ALIAS, OMNIPOST_KEY_PASSWORD. Debug and release both use it so a CI APK installs over a local APK. Do not change the keystore.

A compile failure in Run unit tests looks like a test failure in the Actions UI because that step compiles main + tests together. Missing imports and broken Java fail there in about 1m30s with no APK and no test reports. Real test assertion failures are rarer.

To skip CI on a docs-only push, put [skip ci] in the commit message.

13. Tests

JVM unit tests only (`testOptions.unitTests.isReturnDefaultValues = true`). They must not construct Android framework types or `ScheduleCoordinator` (that class pulls `Context`, `Room`, `FGS`).

scheduler/CaptionTemplatesTest.java

Fill, linger, {today} vs {date}, nested vars, countdown grammar. ASCII only.

scheduler/DateUtilsTest.java

day math, nagFromOrigin, pretty range.

scheduler/RoutineGeneratorTest.java

Keys and windows for weeklies / countdown / notice.

scheduler/TaskSectionsTest.java

Home ordering, snoozed-soonest-first.

scheduler/WeekdaysTest.java

Bitmask.

scheduler/AlarmSchedulerTest.java

requestCode packing (pure).

notify/AlertPlanTest.java

Escalate delays.

share/InstagramStyleTest.java

Bold/italic code points, countdown converts without leftover *.

share/WhatsAppPreviewTest.java

Preview.

util/DeskBackupTest.java

JSON round-trip of captions with quotes/newlines; reject garbage.

14. Find-it index (I want to change X)

Alarm time / 1-minute lead / 5-minute ring

scheduler/ScheduleTimes.java then AlarmScheduler and AlertPlan

What time the evening draft fires

Prefs.draftHour + SettingsFragment.pickDraftHour + RoutineGenerator dayBeforeAt

Vibrate then ring

Prefs.alertMode, AlarmPulse, AlertPlan, Settings row_alert

Stop the ring when any button is tapped

AlarmActivity.acknowledge -> NotificationHelper.hush -> AlarmPulse.silence

Lock-screen alarm UI / title lag

AlarmLaunch extras; AlarmActivity.paintFromIntent before load()

Always-on notification text

NotificationHelper.buildDesk + TaskStatus.nextToRing

Home sort order

TaskSections + TaskStatus.nextRingMillis

Add weekly / daily / countdown / notice

SeriesEditor + ScheduleCoordinator.saveSeries + RoutineGenerator.addSeries

Hardcoded Sunday/Wed/Fri and fasting

RoutineGenerator.generate (not Series)

Beyond Limit dates / seed caption

Campaigns.java + SeriesDefaults.countdownCaption()

Birthday NOTICE month name

CaptionTemplates.monthNameFromOccurrence + SeriesDefaults.noticeCaption

Caption tokens / {today} vs {date}

CaptionVars.builtins + CaptionTemplates.putBuiltins

Linger across countdown days

CaptionTemplates.lingerDraft/sharedTemplate + ScheduleCoordinator.rememberSeriesCaption + DraftActivity persist/load

Save means stop write-caption nags

DraftActivity.persist(true) -> markCaptionSaved; captionSavedAt

Snooze labels as clock times

AlarmActivity.paintNagButtons + DateUtils.nagFromOrigin + DateUtils.prettyClock

WhatsApp package

share/WhatsAppRouter.java

Instagram bold/italic

share/InstagramStyle.java

Export after uninstall

util/DeskBackup.java + SettingsFragment Keep a copy

Permissions / OEM survival

util/SurvivalHelper.java + Settings survival rows + DeskKeepAliveService

Room schema change

entity + AppDatabase version + new Migration + addMigrations

User-visible copy

res/values/strings.xml

App icon

res/mipmap-* and res/drawable/ic_launcher_*

CI APK

.github/workflows/android.yml artifact app-debug

15. Standing rules (do not break these)

- App name OmniPost. Package stays damjay.publicity.omnipost.
- Exact AlarmClock + FGS START_STICKY + BootReceiver. Not WorkManager-only.
- Sound and alarm UI from extras, before Room. Title on AlarmActivity must not wait for load().
- WhatsApp via com.whatsapp only. No system share sheet.
- Keep the provided keystore. CI debug APKs must install over local ones.
- Only explicit Save sets captionSavedAt. onPause and opening a seed draft must not.
- rememberSeriesCaption must skip canned templates so a seed autosave cannot wipe a custom countdown caption.
- Do not restore DraftActivity.load() overwriting variantA with a filled caption.
- Countdown: one card, 3-day generate window, drop past days, linger the user's template, fill {Days}/{away} live.
- Next-up notification and first home row = soonest nextRingMillis, including snooze.
- Writing captions = soft pulse. Sending the flyer = loud + 1-minute lead.
- Tapping any alarm action stops the ring immediately (silent rooms).
- Room migrations must be registered. fallbackToDestructiveMigration is a last resort, not a plan.
- Tests must not call ScheduleCoordinator (Android + Room). Keep linger tests ASCII.
- Do not push changes to .github/workflows unless the repo App permission includes workflows.

If you are lost: start at ScheduleCoordinator.bootstrap, then RoutineGenerator, then AlarmScheduler, then TaskAlarmReceiver -> AlarmLaunch. Everything else hangs off that spine.